

ARTIFICIAL INTELLIGENCE & DEEP LEARNING

Lecture 4: Neural Networks Fundamentals

1. Architecture of Artificial Neural Networks

An Artificial Neural Network (ANN) consists of input layers, one or more hidden layers, and an output layer. Neur

2. Essential Activation Functions

- ReLU (Rectified Linear Unit): Computes $f(x) = \max(0, x)$. It allows deep models to train rapidly and avoids vanishing gradients.
- Sigmoid: Compresses inputs into $[0, 1]$. Best suited for binary classification output probabilities.
- Softmax: Normalizes real numbers into a categorical probability distribution summing to 1.0.

3. Training with Backpropagation and Gradient Descent

- Forward Propagation: Propagates inputs forward through network layers to produce predictions.
- Loss Computation: Measures error via Cross-Entropy Loss or Mean Squared Error.
- Backpropagation: Uses the Chain Rule of calculus to compute partial derivatives (gradients) of the loss function with respect to weights.
- Optimizer: Gradient Descent updates weights in the direction opposite to the gradient: $W = W - \eta * \text{grad}_W$.

4. Preventing Overfitting

- Dropout: Randomly drops units during training to prevent co-adaptation.
- Weight Regularization (L2 / Weight Decay): Penalizes overly large weight coefficients.
- Early Stopping: Halts training when validation performance ceases to improve.